# MATH 616 Midterm Review: Computational Linear Algebra

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Fall 2026

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# 1 Norms, orthogonality, and rank

## 1.1 Vector norms

A norm satisfies $\|x\| \geq 0$ with equality only for $x = 0$, $\|\alpha x\| = |\alpha|\|x\|$, and the triangle inequality $\|x + y\| \leq \|x\| + \|y\|$. The $p$-norms are

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}, \qquad \|x\|_\infty = \max_i |x_i|.$$

All norms on $\mathbb{R}^n$ are equivalent, and

$$\|x\|_\infty \le \|x\|_2 \le \|x\|_1 \le n\|x\|_\infty.$$

The Cauchy–Schwarz inequality is $|x^T y| \le \|x\|_2 \|y\|_2$, with equality exactly when $x$ and $y$ are parallel.

### 1.2 Matrix norms

The norm *induced* (or *subordinate*) by a vector norm is

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}.$$

- $\|A\|_1$ is the maximum absolute column sum.
- $\|A\|_\infty$ is the maximum absolute row sum.
- $\|A\|_2 = \sigma_1(A)$ is the largest singular value; it has no entrywise formula.

Every induced norm satisfies $\|Ax\| \le \|A\|\|x\|$, $\|AB\| \le \|A\|\|B\|$, and $\|I\| = 1$. The Frobenius norm

$$\|A\|_F = \Big(\sum_{i,j} |a_{ij}|^2\Big)^{1/2} = \sqrt{\operatorname{tr}(A^T A)} = \Big(\sum_i \sigma_i^2\Big)^{1/2}$$

is submultiplicative but *not* induced, because $\|I\|_F = \sqrt{n}$. Both $\|\cdot\|_2$ and $\|\cdot\|_F$ are unchanged by multiplication by orthogonal matrices.

### 1.3 Orthogonality, projectors, and rank

A square $Q$ is orthogonal if $Q^T Q = I$. Then $Q^{-1} = Q^T$ and $\|Qx\|_2 = \|x\|_2$, so orthogonal changes of basis never amplify errors. A projector satisfies $P^2 = P$. It is an *orthogonal* projector if also $P^T = P$; for a matrix $\hat{Q}$ with orthonormal columns, $P = \hat{Q}\hat{Q}^T$ projects onto $\operatorname{range}(\hat{Q})$, and $I - P$ projects onto its orthogonal complement.

For $A \in \mathbb{R}^{m \times n}$, $\operatorname{range}(A) = \{Ax\}$, $\operatorname{null}(A) = \{x : Ax = 0\}$, and $\operatorname{rank}(A) = \dim \operatorname{range}(A)$. The rank–nullity theorem says $\operatorname{rank}(A) + \dim \operatorname{null}(A) = n$, and $\operatorname{null}(A) = \operatorname{range}(A^T)^\perp$.

**You should be able to:** compute $\|x\|_{1,2,\infty}$ and $\|A\|_{1,\infty,F}$ by hand; explain why $\|A\|_2$ needs the SVD; find the range, null space, and rank of a small matrix.

## 2 The singular value decomposition and its applications

### 2.1 The theorem

**Theorem 2.1** (SVD)**.** *Every $A \in \mathbb{R}^{m \times n}$ factors as $A = U\Sigma V^T$, with $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ orthogonal and $\Sigma \in \mathbb{R}^{m \times n}$ diagonal with entries $\sigma_1 \ge \sigma_2 \ge \ldots \ge \sigma_{\min(m,n)} \ge 0$. Column by column, $Av_i = \sigma_i u_i$.*

If $\operatorname{rank}(A) = r$, the reduced SVD keeps the first $r$ columns, and

$$A = \hat{U}\hat{\Sigma}\hat{V}^T = \sum_{i=1}^{r} \sigma_i u_i v_i^T.$$

Everything about $A$ can be read off the SVD:

- $\operatorname{rank}(A) = r$, the number of nonzero singular values;
- $\operatorname{range}(A) = \operatorname{span}\{u_1, \dots, u_r\}$ and $\operatorname{null}(A) = \operatorname{span}\{v_{r+1}, \dots, v_n\}$;
- $\|A\|_2 = \sigma_1$ and $\|A\|_F = (\sigma_1^2 + \cdots + \sigma_r^2)^{1/2}$;
- for square invertible $A$, $\kappa_2(A) = \sigma_1/\sigma_n$ and $|\det A| = \sigma_1 \cdots \sigma_n$.

Geometrically, $A$ maps the unit sphere to a hyperellipse with semiaxes $\sigma_i u_i$.

**Computing an SVD by hand.** Since $A^T A = V(\Sigma^T \Sigma)V^T$, the $\sigma_i^2$ are the eigenvalues of $A^T A$ and the $v_i$ are its unit eigenvectors. Then $u_i = Av_i/\sigma_i$ for $\sigma_i > 0$. This is a hand method only: forming $A^T A$ squares the condition number, so it is never used in floating point. Singular values are not eigenvalues in general; for symmetric $A$, $\sigma_i = |\lambda_i|$.

## 2.2 Low-rank approximation

**Theorem 2.2** (Eckart–Young–Mirsky)**.** *Let $A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T$. Among all matrices $B$ of rank at most $k$,*

$$\|A - A_k\|_2 = \sigma_{k+1}, \qquad \|A - A_k\|_F = \left(\sum_{i>k} \sigma_i^2\right)^{1/2}$$

*are the smallest possible errors.*

Storing $A_k$ takes $k(m+n+1)$ numbers instead of $mn$. Fast decay of the singular values means the matrix is approximately low-dimensional; this is the basis of image compression, PCA, recommender systems, and LoRA.

## 2.3 Numerical rank

In floating point, exact rank is meaningless: rounding makes almost every matrix full rank. The $\varepsilon$-rank

$$\operatorname{rank}_\varepsilon(A) = \#\{i : \sigma_i > \varepsilon\}$$

counts singular values above a noise level $\varepsilon$. By Eckart–Young, $\sigma_{k+1}$ is the distance from $A$ to the nearest matrix of rank $k$. Always look for a clear gap in the spectrum; without a gap, the numerical rank is genuinely ambiguous.

## 2.4 The pseudoinverse

**Definition 2.3.** If $A = \hat{U}\hat{\Sigma}\hat{V}^T$ is the reduced SVD, the Moore–Penrose pseudoinverse is

$$A^+ = \hat{V}\hat{\Sigma}^{-1}\hat{U}^T = \sum_{i=1}^{r} \frac{1}{\sigma_i} v_i u_i^T \in \mathbb{R}^{n \times m}.$$

- If $A$ is square and invertible, $A^+ = A^{-1}$.
- If $A$ has full column rank, $A^+ = (A^T A)^{-1} A^T$.
- $x = A^+ b$ is the minimum-norm least squares solution of $Ax = b$.
- $A^+ A$ is the orthogonal projector onto range$(A^T)$, and $AA^+$ onto range$(A)$.

The danger is the factor $1/\sigma_i$: small singular values amplify noise in $b$ enormously. The remedy is to truncate the sum at the numerical rank (truncated SVD) or to regularize.

### 2.5 Principal component analysis

Center the data points $x_1, \dots, x_m \in \mathbb{R}^n$ and stack them as the rows of $X \in \mathbb{R}^{m \times n}$. The sample covariance is $C = \frac{1}{m-1} X^T X$, which is symmetric positive semidefinite. If $X = U\Sigma V^T$, then:

- the principal directions are the columns $v_i$ of $V$;
- the variance along $v_i$ is $\lambda_i(C) = \sigma_i^2/(m-1)$;
- the $k$-dimensional scores are $XV_k = U_k \Sigma_k$;
- the explained variance ratio of $k$ components is $\sum_{i \le k} \sigma_i^2 \big/ \sum_i \sigma_i^2$.

Compute PCA from the SVD of $X$, never by forming $X^T X$.

**You should be able to:** compute the SVD of a $2 \times 2$ matrix by hand; write the best rank-$k$ approximation and its error in both norms; compute a pseudoinverse and a least squares solution; read the numerical rank off a spectrum; explain PCA as an SVD.

## 3 Variational principles and perturbation

Let $S$ be symmetric with eigenvalues $\lambda_1 \ge \dots \ge \lambda_n$ and orthonormal eigenvectors $q_1, \dots, q_n$.

### 3.1 The Rayleigh quotient

$$R(x) = \frac{x^T S x}{x^T x}, \qquad x \neq 0.$$

Writing $x = \sum_i c_i q_i$ gives $R(x) = \sum_i \lambda_i c_i^2 \big/ \sum_i c_i^2$, a weighted average of the eigenvalues. Hence $\lambda_n \le R(x) \le \lambda_1$, with $\lambda_1 = \max R$ and $\lambda_n = \min R$. Near an eigenvector the quotient is flat: an $O(\varepsilon)$ error in the vector gives only an $O(\varepsilon^2)$ error in the eigenvalue estimate.

### 3.2 Courant–Fischer

**Theorem 3.1** (Courant–Fischer)**.**

$$\lambda_k = \max_{\dim \mathcal{V} = k} \; \min_{0 \neq x \in \mathcal{V}} R(x) = \min_{\dim \mathcal{W} = n-k+1} \; \max_{0 \neq x \in \mathcal{W}} R(x).$$

The optimal subspace is $\operatorname{span}\{q_1, \dots, q_k\}$. The proof is a dimension count: any $k$-dimensional subspace meets $\operatorname{span}\{q_k, \dots, q_n\}$ in a nonzero vector, on which $R \le \lambda_k$. For singular values, $\sigma_k(A) = \max_{\dim \mathcal{V}=k} \min_{0 \neq x \in \mathcal{V}} \|Ax\|_2 / \|x\|_2$.

### 3.3 Weyl's inequalities

**Theorem 3.2** (Weyl)**.** *For symmetric $S$ and $E$, $|\lambda_k(S+E) - \lambda_k(S)| \le \|E\|_2$ for every $k$. For any $A, E \in \mathbb{R}^{m \times n}$, $|\sigma_k(A+E) - \sigma_k(A)| \le \|E\|_2$ for every $k$.*

Eigenvalues of symmetric matrices and singular values of all matrices are therefore perfectly conditioned. Consequence for noisy data $\tilde{A} = A + E$ with $\operatorname{rank}(A) = k$ and $\|E\|_2 = \varepsilon$: $\sigma_i(\tilde{A}) \ge \sigma_i(A) - \varepsilon$ for $i \le k$, and $\sigma_i(\tilde{A}) \le \varepsilon$ for $i > k$. A strong signal shows up as $k$ large singular values above a noise floor at level $\varepsilon$.

**You should be able to:** evaluate Rayleigh quotients and bound eigenvalues with them; state Courant–Fischer and Weyl; use Weyl to give intervals for perturbed eigenvalues or singular values.

## 4 Schur form, the spectral theorem, and PSD matrices

### 4.1 Similarity and the Schur form

Similar matrices $B = S^{-1}AS$ share eigenvalues, characteristic polynomial, trace, and determinant. Diagonalization $A = X\Lambda X^{-1}$ can fail (the defective matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has only one eigenvector), and even when it exists, an ill-conditioned $X$ amplifies errors.

**Theorem 4.1** (Schur)**.** *Every $A \in \mathbb{C}^{n \times n}$ can be written $A = QTQ^*$ with $Q$ unitary and $T$ upper triangular. The diagonal of $T$ holds the eigenvalues of $A$.*

The first $k$ Schur vectors span an invariant subspace. A real matrix has a real Schur form with $1 \times 1$ and $2 \times 2$ diagonal blocks, the latter for complex-conjugate pairs. The off-diagonal part of $T$ records nonnormal coupling that eigenvalues alone miss: $\begin{bmatrix} 0 & 1000 \\ 0 & 0 \end{bmatrix}$ has both eigenvalues 0 but stretches $e_2$ by 1000. The QR algorithm computes the Schur form.

### 4.2 Normal matrices and the spectral theorem

$A$ is normal if $A^*A = AA^*$. Normal matrices are exactly those that are unitarily diagonalizable, $A = Q\Lambda Q^*$; their Schur form is diagonal.

**Theorem 4.2** (Spectral theorem)**.** *If $A = A^T$ is real, then $A = Q\Lambda Q^T$ with $Q$ orthogonal and $\Lambda$ real diagonal. The eigenvalues are real and the eigenvectors are orthonormal.*

### 4.3 Positive semidefinite matrices

A symmetric $A$ is positive semidefinite (PSD) if $x^T Ax \geq 0$ for all $x$, and positive definite (PD) if $x^T Ax > 0$ for all $x \neq 0$. For symmetric $A$, the following are equivalent:

1. $A$ is PSD (quadratic-form test);
2. every eigenvalue satisfies $\lambda_i \geq 0$ (eigenvalue test);
3. $A = B^T B$ for some $B$ (factorization test).

For PD, replace $\geq$ by $>$ and require $B$ to have full column rank. Gram matrices $X^T X$ and covariance matrices are always PSD, because $x^T X^T Xx = \|Xx\|_2^2 \geq 0$. A PSD matrix has a square root $A^{1/2} = Q\Lambda^{1/2}Q^T$, and a PD matrix has a Cholesky factorization $A = LL^T$. Nonnegative entries do *not* imply PSD. In floating point a tiny negative eigenvalue, at the level of rounding, usually means a PSD matrix spoiled by roundoff; a large negative eigenvalue means the matrix really is not PSD.

**You should be able to:** explain why Schur form exists when diagonalization fails; diagonalize a symmetric $2 \times 2$ matrix; classify a symmetric matrix as PD, PSD, or indefinite, and prove that a Gram matrix is PSD.

## 5 Gaussian elimination and pivoting

### 5.1 Triangular systems

Backward substitution solves $Ux = b$ from the last equation up, $x_i = \left(b_i - \sum_{j>i} u_{ij}x_j\right)/u_{ii}$; forward substitution solves $Lx = b$ from the top down. Each costs about $n^2/2$ multiplications and as many additions, that is about $n^2$ flops. Both fail only if a diagonal entry is zero, that is, if the triangular matrix is singular.

### 5.2 Gaussian elimination

At step $k$, for each row $j > k$, subtract $\ell_{jk} = a_{jk}^{(k-1)}/a_{kk}^{(k-1)}$ times row $k$, which zeros the entries below the pivot $a_{kk}^{(k-1)}$. After $n-1$ steps the system is upper triangular, and backward substitution finishes the solve. The three elementary row operations (scale, eliminate, interchange) do not change the solution set. Cramer's rule, by contrast, costs $O(n!)$ operations and is never used for computation.

**Cost.** Elimination takes about $n^3/3$ multiplications and $n^3/3$ additions, about $\frac{2}{3}n^3$ flops in total. Doubling $n$ multiplies the work by eight. The substitutions and the pivot search are only $O(n^2)$.

### 5.3 Pivoting

A zero pivot stops naive elimination even when $A$ is invertible. A tiny pivot is worse, because it returns a wrong answer without warning: with $\varepsilon = 10^{-8}$ and four-digit arithmetic, eliminating $\begin{bmatrix} \varepsilon & 1 \\ 2 & 3 \end{bmatrix}$ with pivot $\varepsilon$ loses $x_1$ completely.

- **Partial pivoting.** At step $k$, swap row $k$ with the row $p \geq k$ that has the largest $|a_{pk}|$. Every multiplier then satisfies $|\ell_{jk}| \leq 1$. Extra cost: about $n^2/2$ comparisons.
- **Scaled partial pivoting.** Compute the row scales $s_i = \max_j |a_{ij}|$ once, then choose $p$ to maximize $|a_{pk}|/s_p$. This protects against rows that are scaled very differently, which can fool plain partial pivoting.

### 5.4 Several right-hand sides and the inverse

If all right-hand sides are known in advance, eliminate once on $[A \mid b_1 \ \cdots \ b_m]$ and then do $m$ back substitutions. Computing $A^{-1}$ by solving $AX = I$ costs about $\frac{4}{3}n^3$ multiplications, or $\frac{8}{3}n^3$ flops, roughly four times one solve. To solve $Ax = b$, never form $A^{-1}$: it is slower and less accurate than solving directly or reusing a factorization.

**You should be able to:** carry out elimination with partial or scaled pivoting on a $3 \times 3$ system by hand; count operations; explain why a small pivot is dangerous.

## 6 Matrix factorizations

### 6.1 LU factorization

Step $k$ of elimination is multiplication by the Gauss transformation $M_k = I - \tau^{(k)} e_k^T$, where $\tau^{(k)}$ holds the multipliers below position $k$. Its inverse is $M_k^{-1} = I + \tau^{(k)} e_k^T$. Collecting the steps gives $A = LU$, where $U$ is the reduced matrix and $L$ is unit lower triangular with entries $\ell_{jk}$, exactly the multipliers. Nothing new is computed: the multipliers are simply kept.

**Theorem 6.1** (Existence and uniqueness)**.** *If every leading principal submatrix $A_k$ $(k = 1, \ldots, n)$ is nonsingular, then $A$ has a unique factorization $A = LU$ with $\ell_{ii} = 1$.*

Uniqueness: if $L_1 U_1 = L_2 U_2$, then $L_2^{-1} L_1 = U_2 U_1^{-1}$ is both lower and upper triangular with unit diagonal, so it equals $I$. Doolittle's algorithm normalizes $\ell_{ii} = 1$; Crout's normalizes $u_{ii} = 1$. Once $A = LU$ is known, each new right-hand side costs two triangular solves, $Lz = b$ then $Ux = z$, about $2n^2$ flops instead of $\frac{2}{3}n^3$.

### 6.2 Strictly diagonally dominant matrices

$A$ is strictly diagonally dominant (SDD) if $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ for every row. By Gershgorin's theorem every eigenvalue lies in a disk centered at $a_{ii}$ of radius $\sum_{j \neq i} |a_{ij}|$, and dominance keeps every disk away from 0. So $A$ and every leading submatrix are nonsingular, and $A = LU$ exists. Elimination preserves dominance, so no pivoting is needed.

### 6.3 Symmetric positive definite matrices: Cholesky

**Theorem 6.2** (Cholesky)**.** *A symmetric $A$ is positive definite if and only if $A = LL^T$ for a lower-triangular $L$ with positive diagonal. Equivalently,* $\det A_k > 0$ *for every leading principal submatrix.*

The algorithm, for $i = 1, \dots, n$:

$$\ell_{ii} = \Big(a_{ii} - \sum_{k<i} \ell_{ik}^2\Big)^{1/2}, \qquad \ell_{ji} = \Big(a_{ji} - \sum_{k<i} \ell_{jk}\ell_{ik}\Big)\Big/\ell_{ii}, \quad j > i.$$

It costs about $n^3/6$ multiplications, half of LU, needs no pivoting, and is stable. The square root fails exactly when $A$ is not positive definite, which makes an attempted Cholesky factorization the practical test for positive definiteness. The variant $A = LDL^T$ avoids square roots.

### 6.4 Tridiagonal matrices

If $a_{ij} = 0$ for $|i - j| > 1$, elimination never leaves the three diagonals: $L$ has one subdiagonal and $U$ one superdiagonal, equal to that of $A$. With $u_{11} = a_{11}$, for $i = 1, \dots, n-1$:

$$\ell_{i+1,i} = a_{i+1,i}/u_{ii}, \qquad u_{i+1,i+1} = a_{i+1,i+1} - \ell_{i+1,i}\, a_{i,i+1}.$$

The whole factorization costs $O(n)$. A banded matrix with bandwidths $n_\ell, n_u$ factors in $O(n\, n_\ell n_u)$ without fill outside the band, as long as no rows are interchanged.

### 6.5 Row interchanges: $PA = LU$

**Theorem 6.3.** *For every nonsingular $A$ there are a permutation matrix $P$, a unit lower-triangular $L$, and an upper-triangular $U$ with $PA = LU$.*

The matrix $\begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix}$ has no LU factorization, but its rows swapped do. With partial pivoting, every interchange must also be applied to the multipliers already stored in $L$. To solve $Ax = b$: solve $Lz = Pb$, then $Ux = z$. Also $\det A = (-1)^m u_{11} \cdots u_{nn}$, where $m$ is the number of interchanges.

**You should be able to:** factor a $3 \times 3$ matrix as $LU$, $PA = LU$, or $LL^T$ by hand and use the factors to solve a system; decide from the leading principal minors whether $A = LU$ exists; state the cost of each factorization.

## 7 Conditioning and stability

### 7.1 The condition number

$$\kappa(A) = \|A\|\,\|A^{-1}\| \geq 1, \qquad \kappa_2(A) = \sigma_1/\sigma_n.$$

It is a property of the problem, not of any algorithm.

**Theorem 7.1** (Perturbation bounds)**.** *If $Ax = b$ and $A\tilde{x} = b + \delta b$, then*

$$\frac{\|\tilde{x} - x\|}{\|x\|} \leq \kappa(A)\, \frac{\|\delta b\|}{\|b\|}.$$

*If $(A + \delta A)\tilde{x} = b + \delta b$ with $\|A^{-1}\delta A\| < 1$, then to first order*

$$\frac{\|\tilde{x} - x\|}{\|x\|} \lesssim \kappa(A)\Big(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta b\|}{\|b\|}\Big).$$

Proof of the first bound: $\tilde{x} - x = A^{-1}\delta b$ gives $\|\tilde{x} - x\| \le \|A^{-1}\|\|\delta b\|$, and $\|b\| \le \|A\|\|x\|$ gives $1/\|x\| \le \|A\|/\|b\|$. Multiply the two.

**A small residual proves nothing on its own.** With $r = b - A\tilde{x}$,

$$\frac{\|x - \tilde{x}\|}{\|x\|} \le \kappa(A)\frac{\|r\|}{\|b\|},$$

so a tiny residual guarantees a tiny error only when $\kappa(A)$ is moderate. Rule of thumb: with $\kappa(A) \approx 10^q$ and working precision $10^{-d}$, expect about $d - q$ correct digits.

### 7.2 Iterative refinement

Given computed factors $B = \tilde{L}\tilde{U} \approx A$: solve $Bx^{(0)} = b$; then repeat $r = b - Ax^{(k)}$ (in higher precision), solve $Bc = r$, and set $x^{(k+1)} = x^{(k)} + c$. This is a fixed-point iteration with matrix $I - B^{-1}A$, and it converges when $\|A^{-1}(B - A)\| < \frac{1}{2}$. Each step costs only $O(n^2)$, and with $\kappa_\infty(A) = 10^q$ and machine precision $10^{-d}$, the iterate $x^{(k)}$ carries about $\min\big(d, (k+1)(d-q)\big)$ correct digits.

### 7.3 Backward stability

An algorithm is *backward stable* if its computed answer is the exact answer for a nearby problem: $(A + \delta A)\tilde{x} = b$ with $\|\delta A\|/\|A\|$ a modest multiple of machine precision. Backward stability plus the perturbation bound gives a forward error of about $\kappa(A)$ times machine precision.

The example $A = \begin{bmatrix} \varepsilon & 1 \\ 1 & 1 \end{bmatrix}$ has $\kappa_\infty(A) \approx 4$, so it is well conditioned. Without pivoting, the computed factors reproduce a matrix that differs from $A$ by $O(1)$, which is unstable. With partial pivoting, the factors are exact for a matrix within $\varepsilon$ of $A$, which is backward stable. Conditioning describes the problem; stability describes the algorithm.

Wilkinson's analysis of partial pivoting gives

$$\frac{\|x - \tilde{x}\|}{\|x\|} \lesssim 4n^2\, g_n(A)\, \kappa_\infty(A)\, \mathrm{Eps}, \qquad g_n(A) = \frac{\max_{i,j,k}|a_{ij}^{(k)}|}{\max_{i,j}|a_{ij}|} \le 2^{n-1},$$

where Eps is machine precision. The growth factor $g_n$ can reach $2^{n-1}$ for special matrices, but it stays small for matrices met in practice, which is why partial pivoting is the standard.

**You should be able to:** compute $\kappa_1$ or $\kappa_\infty$ of a $2 \times 2$ matrix; apply both perturbation bounds; explain the difference between an ill-conditioned problem and an unstable algorithm.

## 8 Iterative methods

### 8.1 The framework

For large sparse systems, $O(n^3)$ work and the fill-in of $L$ and $U$ are unaffordable. Choose an invertible $Q$ that is easy to solve with and close to $A$, and iterate

$$x^{(k+1)} = Mx^{(k)} + d, \qquad M = I - Q^{-1}A, \qquad d = Q^{-1}b,$$

or equivalently $x^{(k+1)} = x^{(k)} + Q^{-1}(b - Ax^{(k)})$. If the iteration converges, its limit solves $Ax = b$. One sweep costs one product with $A$ plus one solve with $Q$: about $2\,\mathrm{nnz}(A)$ flops.

## 8.2 The three methods

Split $A = D - L - U$, with $D$ the diagonal and $-L$, $-U$ the strictly lower and upper parts. Component by component:

$$x_i^{(k+1)} = \frac{1}{a_{ii}}\Big(b_i - \sum_{j<i} a_{ij} x_j^{(\star)} - \sum_{j>i} a_{ij} x_j^{(k)}\Big),$$

where Jacobi uses $\star = k$ (old values only) and Gauss–Seidel uses $\star = k+1$ (new values as soon as they exist). SOR blends: $x_i^{(k+1)} = \omega\, x_i^{\mathrm{GS}} + (1-\omega) x_i^{(k)}$.

Table 1: Splittings and iteration matrices of the three methods

| Method | $Q$ | Iteration matrix $M$ |
|---|---|---|
| Jacobi | $D$ | $M_{\mathrm{J}} = D^{-1}(L+U)$ |
| Gauss–Seidel | $D - L$ | $M_{\mathrm{GS}} = (D-L)^{-1}U$ |
| SOR | $\frac{1}{\omega}D - L$ | $M_{\mathrm{SOR}} = (D-\omega L)^{-1}\big[(1-\omega)D + \omega U\big]$ |

Jacobi parallelizes perfectly; Gauss–Seidel uses one vector and is usually faster, but its sweep is sequential and depends on the ordering. SOR with $\omega = 1$ is Gauss–Seidel; $\omega > 1$ is over-relaxation.

## 8.3 Convergence

The error $e^{(k)} = x^{(k)} - x^*$ satisfies $e^{(k)} = M^k e^{(0)}$. The spectral radius is $\rho(M) = \max_i |\lambda_i(M)|$.

**Theorem 8.1** (Sufficient condition)**.** *If $\|M\| < 1$ in some induced norm, the iteration converges from every $x^{(0)}$, and*

$$\|x^{(k)} - x^*\| \le \frac{\|M\|^k}{1 - \|M\|}\|x^{(1)} - x^{(0)}\|.$$

**Theorem 8.2.** *For every square $A$, $\rho(A) = \inf\|A\|$ over all induced norms. In particular $\rho(A) \le \|A\|$ for every induced norm.*

**Theorem 8.3** (Convergence theorem)**.** *The following are equivalent: (1) $x^{(k+1)} = Mx^{(k)} + d$ converges for every $x^{(0)}$; (2) $\rho(M) < 1$; (3) $\|M^k\| \to 0$.*

The norm condition is sufficient but depends on the norm; $\rho(M) < 1$ is necessary and sufficient. Each sweep multiplies the error by about $\rho(M)$, so gaining $p$ digits takes about $p/(-\log_{10}\rho(M))$ sweeps.

### 8.4 Special matrices

- **SDD.** Jacobi and Gauss–Seidel converge for every start. For Jacobi,

$$\|M_{\mathrm{J}}\|_\infty = \max_i \frac{1}{|a_{ii}|} \sum_{j \neq i} |a_{ij}| < 1.$$

  For Gauss–Seidel, an eigenvector argument gives $\rho(M_{\mathrm{GS}}) < 1$.

- **Stein–Rosenberg.** If $a_{ii} > 0$ and $a_{ij} \leq 0$ for $i \neq j$, then Jacobi and Gauss–Seidel converge or diverge together, and when they converge, Gauss–Seidel is faster.

- **Kahan.** $\rho(M_{\mathrm{SOR}}(\omega)) \geq |\omega - 1|$ for every $\omega$, because $\det M_{\mathrm{SOR}} = (1-\omega)^n$. SOR can only converge for $0 < \omega < 2$.

- **Ostrowski–Reich.** If $A$ is symmetric positive definite, SOR converges for every $0 < \omega < 2$; in particular Gauss–Seidel always converges. Jacobi need not.

- **SPD tridiagonal.** $\rho(M_{\mathrm{GS}}) = \rho(M_{\mathrm{J}})^2$, and the optimal parameter is

$$\omega_{\mathrm{opt}} = \frac{2}{1 + \sqrt{1 - \rho(M_{\mathrm{J}})^2}}, \qquad \rho\Big(M_{\mathrm{SOR}}(\omega_{\mathrm{opt}})\Big) = \omega_{\mathrm{opt}} - 1.$$

### 8.5 When is iteration cheaper?

Total cost is the number of sweeps times about $2\,\mathrm{nnz}(A)$. Iteration wins for very large sparse systems, especially from two- and three-dimensional problems where direct factors fill in. It loses for small or dense systems, for many right-hand sides with the same $A$ (factor once instead), and for tridiagonal systems, which a direct solver handles in $O(n)$. On fine grids $\rho(M_{\mathrm{J}})$ approaches 1, so the number of sweeps grows with $n$; a good $\omega$ in SOR reduces it from order $n$ to order $\sqrt{n}$ for the two-dimensional Poisson problem.

**You should be able to:** perform one sweep of each method by hand; write $M_{\mathrm{J}}$ and $M_{\mathrm{GS}}$ for a small matrix; decide convergence from a norm, the spectral radius, or a structural theorem; compute $\omega_{\mathrm{opt}}$; estimate the number of sweeps.

## 9 Summary of costs

Costs are leading-order flop counts for a dense $n \times n$ matrix, where one flop is one addition or one multiplication.

## 10 Common mistakes

1. Treating singular values as eigenvalues. They agree, up to sign, only for symmetric (more generally, normal) matrices.

Table 2: Leading-order costs of the main algorithms

| Task | Cost |
|---|---|
| Triangular solve (forward or backward) | $n^2$ |
| Gaussian elimination, or LU factorization | $\frac{2}{3}n^3$ |
| Each further solve with known $L$ and $U$ | $2n^2$ |
| Cholesky factorization | $\frac{1}{3}n^3$ |
| Tridiagonal factorization and solve | $O(n)$ |
| Banded factorization, bandwidths $n_\ell$, $n_u$ | $O(n\, n_\ell n_u)$ |
| Explicit inverse from $[A \mid I]$ | $\frac{8}{3}n^3$ |
| Partial pivoting (comparisons) | $O(n^2)$ |
| One Jacobi, Gauss–Seidel, or SOR sweep | $2\,\mathrm{nnz}(A)$ |
| Cramer's rule by cofactor expansion | $O(n!)$ |

2. Computing an SVD or PCA in floating point by forming $A^TA$. It squares the condition number.

3. Calling a matrix PSD because its entries are nonnegative, or forgetting that PSD requires symmetry.

4. Forgetting to swap the stored multipliers in $L$ when rows are interchanged in $PA = LU$.

5. Solving $Ax = b$ by forming $A^{-1}$.

6. Concluding that $\tilde{x}$ is accurate because the residual $b - A\tilde{x}$ is small, without checking $\kappa(A)$.

7. Blaming the matrix for an unstable algorithm, or the algorithm for an ill-conditioned matrix.

8. Concluding that an iteration diverges because $\|M\| \geq 1$ in one norm. Only $\rho(M) \geq 1$ proves divergence for some starting vector.

9. Using SOR with $\omega \leq 0$ or $\omega \geq 2$.

## 11 Practice problems

**Problem 1** (Norms)**.** Let $x = (3, -4, 12)^T$ and $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$. Compute $\|x\|_1$, $\|x\|_2$, $\|x\|_\infty$, $\|A\|_1$, $\|A\|_\infty$, and $\|A\|_F$. Without computing it, explain why $\|A\|_2 \leq \|A\|_F$.

**Problem 2** (SVD by hand)**.** Let $A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$. (a) Compute the SVD of $A$. (b) Read off $\|A\|_2$, $\|A\|_F$, and $\kappa_2(A)$. (c) Find the best rank-one approximation $A_1$ and the error $\|A - A_1\|_2$.

**Problem 3** (Pseudoinverse and least squares)**.** Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 2}, \qquad b = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

(a) Compute the reduced SVD $A = \hat{U}\hat{\Sigma}\hat{V}^T$, with $\hat{U} \in \mathbb{R}^{3\times 2}$. (b) Compute the pseudoinverse $A^+ = (A^TA)^{-1}A^T$, and check that it agrees with $\sum_i \frac{1}{\sigma_i} v_i u_i^T$ in one entry of your choice. (c) Find the least squares solution $x = A^+b$, compute the residual $r = b - Ax$, and verify that $A^Tr = 0$. Explain why the residual of a least squares solution is always orthogonal to range$(A)$.

**Problem 4** (Numerical rank and Weyl)**.** A matrix $A$ has singular values 10, 5, $10^{-3}$, $10^{-4}$. (a) What is the distance in the 2-norm from $A$ to the nearest matrix of rank two, and of rank one? (b) What is $\operatorname{rank}_\varepsilon(A)$ for $\varepsilon = 10^{-2}$? (c) If $\|E\|_2 \le 10^{-2}$, in what intervals do the singular values of $A + E$ lie?

**Problem 5** (Rayleigh quotient and Weyl)**.** Let $S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. (a) Find the eigenvalues and orthonormal eigenvectors of $S$ and write $S = Q\Lambda Q^T$. (b) Compute $R(x)$ for $x = (1, 0)^T$ and check that it lies between the eigenvalues. (c) Without computing them, give intervals for the eigenvalues of $S + E$ with $E = \begin{bmatrix} 0.1 & 0 \\ 0 & 0 \end{bmatrix}$.

**Problem 6** (PSD classification)**.** Classify each matrix as positive definite, positive semidefinite but singular, or indefinite:

$$A_1 = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}, \qquad A_3 = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}.$$

Then prove that $X^TX$ is PSD for every real matrix $X$.

**Problem 7** (Schur versus diagonalization)**.** Explain why $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ cannot be diagonalized, and give a Schur decomposition of $A$.

**Problem 8** ($PA = LU$ by hand)**.** Let

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 2 & 3 \\ -1 & -3 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 4 \\ 7 \\ -4 \end{bmatrix}.$$

(a) Compute $PA = LU$ using partial pivoting. (b) Use the factors to solve $Ax = b$. (c) Compute $\det A$ from the factors.

**Problem 9** (Cholesky)**.** Let $A = \begin{bmatrix} 4 & 2 & -2 \\ 2 & 10 & 2 \\ -2 & 2 & 6 \end{bmatrix}$. (a) Compute the Cholesky factor $L$. (b) Solve $Ax = (4, 14, 6)^T$ using $L$. (c) Explain why this proves that $A$ is positive definite.

**Problem 10** (Conditioning)**.** Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 1.001 \end{bmatrix}$. (a) Compute $\kappa_\infty(A)$. (b) If $b$ is known to relative accuracy $10^{-5}$ in the $\infty$-norm, how large can the relative error in $x$ be? (c) A computed $\tilde{x}$ has relative residual $\|b - A\tilde{x}\|_\infty / \|b\|_\infty = 10^{-8}$. What can you conclude?

**Problem 11** (A tiny pivot)**.** Consider the system

$$\begin{bmatrix} \varepsilon & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1+\varepsilon \\ 5 \end{bmatrix}, \qquad \varepsilon = 10^{-8},$$

whose exact solution is $x = (1,1)^T$. (a) Compute $\kappa_\infty$ of the matrix. Is the problem well conditioned? (b) Solve the system by Gaussian elimination *without* pivoting, rounding every result to four significant digits. (c) Solve it again with partial pivoting, using the same four-digit arithmetic. (d) Explain the difference between (b) and (c) using the words *conditioning* and *stability*.

**Problem 12** (Costs)**.** (a) An LU factorization of a dense $1000 \times 1000$ matrix takes 0.05 seconds. Estimate the time for $n = 4000$. (b) You must solve $Ax = b_j$ for 100 right-hand sides that arrive one at a time. Compare the cost of repeated Gaussian elimination with the cost of one LU factorization followed by triangular solves. (c) Why should you not compute $A^{-1}$ to solve these systems?

**Problem 13** (Jacobi, Gauss–Seidel, SOR)**.** Let

$$A = \begin{bmatrix} 5 & 3 & 0 \\ 3 & 10 & 3 \\ 0 & 3 & 5 \end{bmatrix}, \qquad b = \begin{bmatrix} 8 \\ 16 \\ 8 \end{bmatrix}, \qquad x^* = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

(a) Starting from $x^{(0)} = 0$, perform one Jacobi sweep and one Gauss–Seidel sweep. (b) Write $M_{\mathrm{J}}$ and compute $\|M_{\mathrm{J}}\|_\infty$. What does it tell you? (c) Compute the eigenvalues of $M_{\mathrm{J}}$ and hence $\rho(M_{\mathrm{J}})$. Then, using that $A$ is symmetric positive definite and tridiagonal, find $\rho(M_{\mathrm{GS}})$, $\omega_{\mathrm{opt}}$, and $\rho(M_{\mathrm{SOR}}(\omega_{\mathrm{opt}}))$. (d) Rank the three methods by speed. About how many Jacobi sweeps reduce the error as much as one Gauss–Seidel sweep?

**Problem 14** (Norm versus spectral radius)**.** Let $M = \begin{bmatrix} 0.5 & 10 \\ 0 & 0.5 \end{bmatrix}$. (a) Compute $\|M\|_\infty$ and $\rho(M)$. (b) Does $x^{(k+1)} = Mx^{(k)} + d$ converge for every $x^{(0)}$? Justify. (c) What does Kahan's theorem say about SOR with $\omega = 2.5$?

**Problem 15** (True or false)**.** Justify each answer in one or two sentences.

1. If $A$ is invertible, then $A = LU$ with $\ell_{ii} = 1$ exists.
2. Gaussian elimination with partial pivoting never needs a multiplier larger than 1 in magnitude.
3. A small residual always means a small error.
4. The singular values of $A$ and $A^T$ are the same.
5. If $A$ is symmetric positive definite, Jacobi converges.
6. Gaussian elimination with partial pivoting is backward stable for every matrix, with a backward error bounded independently of $n$.

# 12 Solutions

**Problem 1.** $\|x\|_1 = 19$, $\|x\|_2 = \sqrt{9+16+144} = 13$, $\|x\|_\infty = 12$. Column sums of $|A|$ are $4, 6$, so $\|A\|_1 = 6$; row sums are $3, 7$, so $\|A\|_\infty = 7$; $\|A\|_F = \sqrt{1+4+9+16} = \sqrt{30}$. Since $\|A\|_2^2 = \sigma_1^2 \le \sum_i \sigma_i^2 = \|A\|_F^2$, we have $\|A\|_2 \le \|A\|_F$. (A computer gives $\|A\|_2 \approx 5.12$; you are not expected to compute $\|A\|_2$ by hand.)

**Problem 2.** (a) First

$$A^TA = \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}.$$

Its characteristic polynomial is $(5-\lambda)^2 - 9 = 0$, so $5-\lambda = \pm 3$ and $\lambda = 8$ or $\lambda = 2$. Hence $\sigma_1 = \sqrt{8} = 2\sqrt{2}$ and $\sigma_2 = \sqrt{2}$. For $\lambda = 8$, $A^TA - 8I = \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix}$ gives $v_1 = \frac{1}{\sqrt{2}}(1,1)^T$; for $\lambda = 2$, $A^TA - 2I = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$ gives $v_2 = \frac{1}{\sqrt{2}}(1,-1)^T$. Then

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{2\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad u_2 = \frac{Av_2}{\sigma_2} = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix},$$

so

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

(b) $\|A\|_2 = 2\sqrt{2}$, $\|A\|_F = \sqrt{8+2} = \sqrt{10}$, $\kappa_2(A) = 2$. (c) $A_1 = \sigma_1 u_1 v_1^T = \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}$ and $\|A - A_1\|_2 = \sigma_2 = \sqrt{2}$.

**Problem 3.** (a) The columns of $A$ are $a_1 = (1,0,1)^T$ and $a_2 = (0,1,1)^T$, with $a_1^Ta_1 = 2$, $a_1^Ta_2 = 1$, $a_2^Ta_2 = 2$, so

$$A^TA = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its characteristic polynomial is $(2-\lambda)^2 - 1 = 0$, so $\lambda = 3$ or $\lambda = 1$, and $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$. For $\lambda = 3$, $A^TA - 3I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ gives $v_1 = \frac{1}{\sqrt{2}}(1,1)^T$; for $\lambda = 1$, $A^TA - I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ gives $v_2 = \frac{1}{\sqrt{2}}(1,-1)^T$. Then

$$u_1 = \frac{Av_1}{\sigma_1} = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \qquad u_2 = \frac{Av_2}{\sigma_2} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

So $\hat{U} = [u_1\ u_2]$ is $3 \times 2$, $\hat{\Sigma} = \mathrm{diag}(\sqrt{3}, 1)$, and $\hat{V} = [v_1\ v_2]$. Only two left singular vectors are needed, because $A$ has only two columns. (b) The inverse of a $2 \times 2$ matrix is $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, so

$$(A^TA)^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \qquad A^+ = (A^TA)^{-1}A^T = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \end{bmatrix}.$$

Check the $(1,1)$ entry with the SVD formula: $\frac{1}{\sqrt{3}}(v_1)_1(u_1)_1 + \frac{1}{1}(v_2)_1(u_2)_1 = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{6}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{6} + \frac{1}{2} = \frac{2}{3}$, which matches. (c) $x = A^+b = \frac{1}{3}(2-2+0,\ -1+4+0)^T = (0,1)^T$. Then $Ax = (0,1,1)^T$ and $r = b - Ax = (1,1,-1)^T$, so $A^Tr = (1+0-1,\ 0+1-1)^T = (0,0)^T$. The least squares

solution minimizes $\|b - Ax\|_2$, so $Ax$ is the orthogonal projection of $b$ onto range$(A)$, and the residual is perpendicular to that subspace. Algebraically, $A^T r = 0$ is exactly the normal equations $A^T Ax = A^T b$.

**Problem 4.** (a) By Eckart–Young the distances are $\sigma_3 = 10^{-3}$ to rank two and $\sigma_2 = 5$ to rank one. (b) Two singular values exceed $10^{-2}$, so $\operatorname{rank}_\varepsilon(A) = 2$. (c) By Weyl each singular value moves by at most $10^{-2}$: $\sigma_1 \in [9.99, 10.01]$, $\sigma_2 \in [4.99, 5.01]$, $\sigma_3 \in [0, 0.011]$, $\sigma_4 \in [0, 0.0101]$.

**Problem 5.** (a) The characteristic polynomial is $(2-\lambda)^2 - 1 = 0$, so $2 - \lambda = \pm 1$, giving $\lambda_1 = 3$ and $\lambda_2 = 1$. For $\lambda_1 = 3$, $S - 3I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$ forces $x_1 = x_2$, so $q_1 = \frac{1}{\sqrt{2}}(1,1)^T$. For $\lambda_2 = 1$, $S - I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ forces $x_1 = -x_2$, so $q_2 = \frac{1}{\sqrt{2}}(1,-1)^T$. Then $Q = [q_1 \; q_2]$ and $\Lambda = \operatorname{diag}(3,1)$. (b) $R(e_1) = s_{11} = 2 \in [1,3]$. (c) $\|E\|_2 = 0.1$, so the eigenvalues of $S + E$ lie in $[2.9, 3.1]$ and $[0.9, 1.1]$. (A computer gives about 3.051 and 1.049, inside both intervals; Weyl's theorem gives the intervals without computing them.)

**Problem 6.** For a symmetric $2 \times 2$ matrix the eigenvalues are the roots of $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$. $A_1$: tr $= 4$, det $= 3$, so $\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$ and the eigenvalues are $3, 1$: positive definite. $A_2$: tr $= 2$, det $= 1 - 9 = -8$, so $\lambda^2 - 2\lambda - 8 = (\lambda - 4)(\lambda + 2)$ and the eigenvalues are $4, -2$: indefinite. For example $x = (1,-1)^T$ gives $x^T A_2 x = 1 - 6 + 1 = -4 < 0$. $A_3$: tr $= 5$, det $= 4 - 4 = 0$, so $\lambda^2 - 5\lambda = \lambda(\lambda - 5)$ and the eigenvalues are $5, 0$: PSD and singular. Indeed $x^T A_3 x = 4x_1^2 + 4x_1 x_2 + x_2^2 = (2x_1 + x_2)^2 \geq 0$, with equality for $x = (1,-2)^T$. For any $X$ and $x$, $x^T X^T X x = (Xx)^T(Xx) = \|Xx\|_2^2 \geq 0$, and $X^T X$ is symmetric.

**Problem 7.** The only eigenvalue is 1, and $A - I = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$ has a one-dimensional null space spanned by $e_1$. There is no basis of eigenvectors, so $A = X\Lambda X^{-1}$ is impossible. $A$ is already upper triangular, so $Q = I$, $T = A$ is a Schur decomposition.

**Problem 8.** (a) *Column 1.* The largest entry in absolute value is 2, in row 2, so swap rows 1 and 2. The multipliers are $\ell_{21} = 1/2$ and $\ell_{31} = -1/2$. Row 2 becomes $(1,2,1) - \frac{1}{2}(2,2,3) = (0, 1, -\frac{1}{2})$, and row 3 becomes $(-1,-3,0) + \frac{1}{2}(2,2,3) = (0, -2, \frac{3}{2})$:

$$\begin{bmatrix} 2 & 2 & 3 \\ 0 & 1 & -\frac{1}{2} \\ 0 & -2 & \frac{3}{2} \end{bmatrix}.$$

*Column 2.* $|-2| > |1|$, so swap rows 2 and 3, and swap the stored multipliers $\frac{1}{2}$ and $-\frac{1}{2}$ with them. The new multiplier is $\ell_{32} = 1/(-2) = -\frac{1}{2}$, and row 3 becomes $(0, 1, -\frac{1}{2}) + \frac{1}{2}(0, -2, \frac{3}{2}) = (0, 0, \frac{1}{4})$. This gives

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 2 & 3 \\ 0 & -2 & \frac{3}{2} \\ 0 & 0 & \frac{1}{4} \end{bmatrix}.$$

(b) $Pb = (7, -4, 4)^T$. Forward substitution gives $z = (7, -\frac{1}{2}, \frac{1}{4})^T$, and backward substitution gives $x = (1,1,1)^T$. (c) Two interchanges were made, so $\det A = (-1)^2 \cdot 2 \cdot (-2) \cdot \frac{1}{4} = -1$.

**Problem 9.** (a) $\ell_{11} = 2$, $\ell_{21} = 1$, $\ell_{31} = -1$; $\ell_{22} = \sqrt{10 - 1} = 3$, $\ell_{32} = (2 - (1)(-1))/3 = 1$;

$\ell_{33} = \sqrt{6-1-1} = 2$:

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ -1 & 1 & 2 \end{bmatrix}.$$

(b) $Lz = b$ gives $z = (2, 4, 2)^T$, and $L^T x = z$ gives $x = (1, 1, 1)^T$. (c) Every square root was of a positive number, so $A = LL^T$ with $L$ invertible, and $x^T Ax = \|L^T x\|_2^2 > 0$ for $x \neq 0$.

**Problem 10.** (a) $\det A = 0.001$ and $A^{-1} = \begin{bmatrix} 1001 & -1000 \\ -1000 & 1000 \end{bmatrix}$. So $\|A\|_\infty = 2.001$, $\|A^{-1}\|_\infty = 2001$, and $\kappa_\infty(A) \approx 4004$. (b) Up to $4004 \times 10^{-5} \approx 0.04$, a 4% relative error. (c) Only that the relative error is at most about $4004 \times 10^{-8} \approx 4 \times 10^{-5}$; the residual is about four thousand times too optimistic.

**Problem 11.** (a) $\|A\|_\infty = 5$. Since $\det A = 3\varepsilon - 2 \approx -2$, $A^{-1} \approx -\frac{1}{2}\begin{bmatrix} 3 & -1 \\ -2 & \varepsilon \end{bmatrix}$ and $\|A^{-1}\|_\infty \approx 2$, so $\kappa_\infty(A) \approx 10$. The problem is well conditioned. (b) The multiplier is $2/\varepsilon = 2 \times 10^8$. The new $(2,2)$ entry is $3 - 2 \times 10^8$, which rounds to $-2.000 \times 10^8$, and the new right-hand side $5 - 2 \times 10^8(1+\varepsilon)$ also rounds to $-2.000 \times 10^8$. Back substitution gives $x_2 = 1$, and then $x_1 = \big((1+\varepsilon) - x_2\big)/\varepsilon$, where $1+\varepsilon$ rounds to 1, so $x_1 = 0$. The computed solution $(0, 1)^T$ has a completely wrong first component, with no warning. (c) Partial pivoting swaps the rows, since $|2| > |\varepsilon|$. The multiplier is $\varepsilon/2 = 5 \times 10^{-9}$; the new $(2,2)$ entry $1 - \frac{3}{2}\varepsilon$ and right-hand side $1 + \varepsilon - \frac{5}{2}\varepsilon$ both round to 1. So $x_2 = 1$ and $x_1 = (5-3)/2 = 1$: the correct solution. (d) The problem is well conditioned ($\kappa \approx 10$), so the failure in (b) is not the matrix's fault. Elimination without pivoting is *unstable* here: the huge multiplier $1/\varepsilon$ swamps the original entries, rounding destroys the information they carried, and the computed factors belong to a matrix far from $A$. Partial pivoting keeps every multiplier at most 1 in magnitude, so it is backward stable, and a backward stable algorithm on a well-conditioned problem gives an accurate answer.

**Problem 12.** (a) The cost grows like $n^3$, so a factor 4 in $n$ is a factor 64 in time: about 3.2 seconds. (b) Repeated elimination costs about $100 \cdot \frac{2}{3}n^3$ flops. One LU plus 100 pairs of triangular solves costs about $\frac{2}{3}n^3 + 100 \cdot 2n^2$, which for $n = 1000$ is roughly $0.9 \times 10^9$ against $6.7 \times 10^{10}$, about 75 times cheaper. (c) Forming $A^{-1}$ costs about $\frac{8}{3}n^3$ flops, four times the LU factorization, and each product $A^{-1}b_j$ still costs $2n^2$, the same as the two triangular solves. It is also less accurate than solving with the factors.

**Problem 13.** (a) The equations solved for their own unknowns are

$$x_1 = \frac{8 - 3x_2}{5}, \qquad x_2 = \frac{16 - 3x_1 - 3x_3}{10}, \qquad x_3 = \frac{8 - 3x_2}{5}.$$

Jacobi puts $x^{(0)} = 0$ into every right-hand side: $x^{(1)} = (\frac{8}{5},\ \frac{16}{10},\ \frac{8}{5}) = (\frac{8}{5}, \frac{8}{5}, \frac{8}{5})$. Gauss–Seidel uses each new value at once:

$$x_1 = \frac{8-0}{5} = \frac{8}{5}, \qquad x_2 = \frac{16 - 3 \cdot \frac{8}{5} - 0}{10} = \frac{80-24}{50} = \frac{28}{25}, \qquad x_3 = \frac{8 - 3 \cdot \frac{28}{25}}{5} = \frac{200-84}{125} = \frac{116}{125}.$$

(b) $D = \operatorname{diag}(5, 10, 5)$, and $L + U$ holds the off-diagonal entries of $A$ with their signs changed, so

$$M_{\mathrm{J}} = D^{-1}(L+U) = \begin{bmatrix} 0 & -\frac{3}{5} & 0 \\ -\frac{3}{10} & 0 & -\frac{3}{10} \\ 0 & -\frac{3}{5} & 0 \end{bmatrix}.$$

The absolute row sums are $\frac{3}{5}$, $\frac{3}{10}+\frac{3}{10}=\frac{3}{5}$, and $\frac{3}{5}$, so $\|M_{\mathrm{J}}\|_\infty=\frac{3}{5}<1$. Jacobi therefore converges from every starting vector, and each sweep reduces the error in the $\infty$-norm at least by the factor $\frac{3}{5}$. (This agrees with $A$ being strictly diagonally dominant.) (c) Expanding $\det(\lambda I-M_{\mathrm{J}})$ along the first row,

$$\det\begin{bmatrix}\lambda & \frac{3}{5} & 0\\ \frac{3}{10} & \lambda & \frac{3}{10}\\ 0 & \frac{3}{5} & \lambda\end{bmatrix}=\lambda\Big(\lambda^2-\tfrac{9}{50}\Big)-\tfrac{3}{5}\cdot\tfrac{3}{10}\lambda=\lambda\Big(\lambda^2-\tfrac{9}{50}-\tfrac{9}{50}\Big)=\lambda\Big(\lambda^2-\tfrac{9}{25}\Big).$$

The eigenvalues are $0$ and $\pm\frac{3}{5}$, so $\rho(M_{\mathrm{J}})=\frac{3}{5}$. Since $A$ is symmetric positive definite and tridiagonal,

$$\rho(M_{\mathrm{GS}})=\rho(M_{\mathrm{J}})^2=\tfrac{9}{25},\qquad \omega_{\mathrm{opt}}=\frac{2}{1+\sqrt{1-\frac{9}{25}}}=\frac{2}{1+\frac{4}{5}}=\frac{10}{9},$$

$$\rho\Big(M_{\mathrm{SOR}}(\omega_{\mathrm{opt}})\Big)=\omega_{\mathrm{opt}}-1=\frac{1}{9}.$$

(d) Each sweep multiplies the error by about $\rho$: 0.6 for Jacobi, 0.36 for Gauss–Seidel, and 0.11 for optimal SOR, so SOR is fastest and Jacobi slowest. Because $\rho(M_{\mathrm{GS}})=\rho(M_{\mathrm{J}})^2$, one Gauss–Seidel sweep reduces the error about as much as two Jacobi sweeps.

**Problem 14.** (a) $\|M\|_\infty=10.5$, and $M$ is triangular with eigenvalues $0.5, 0.5$, so $\rho(M)=0.5$. (b) Yes. By the convergence theorem, $\rho(M)<1$ is necessary and sufficient. A norm larger than 1 proves nothing; the error may grow for a few steps before it decays. (c) $\rho(M_{\mathrm{SOR}}(2.5))\geq|2.5-1|=1.5>1$, so SOR with $\omega=2.5$ diverges for some starting vector.

**Problem 15.**

1. False. $\begin{bmatrix}0 & 2\\ 3 & 1\end{bmatrix}$ is invertible, but its first leading principal minor is zero. A permutation is needed: $PA=LU$ always exists.
2. True. The pivot is the largest entry in its column, so every multiplier $|a_{jk}/a_{kk}|\leq 1$.
3. False. The relative error can be as large as $\kappa(A)$ times the relative residual.
4. True. $A^T=V\Sigma^TU^T$ is an SVD of $A^T$ with the same nonzero singular values.
5. False. The SPD matrix with 1 on the diagonal and 0.6 everywhere else has $\rho(M_{\mathrm{J}})=1.2$. Gauss–Seidel does converge for every SPD matrix.
6. False. The backward error bound contains the growth factor $g_n(A)$, which can reach $2^{n-1}$ (Wilkinson's example), so in the worst case almost every digit can be lost for $n\approx 60$ in double precision. For the matrices met in practice the growth factor stays small, which is why partial pivoting is still the standard choice.